

Forensic Science

for High Schools

Book 1

Jeannette Jolley and John Powrie



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Forensic Science

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Emerald City Books

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Forensic Science Book 1 deals with the various fields that make up the study of forensic science. This book introduces the concept of the crime scene and how the crime scene is treated. It continues to follow what happens to any evidence found, explaining the techniques used to investigate evidence. Topic 1 introduces the basic concepts of forensic science, while Topic 2 covers trace evidence and the uses of microscopes, UV light and flame colours. Forensic Science Book 2 examines fingerprinting and forgery in detail.

Each book contains factual information on several distinct subject areas which is interspersed with interesting practical activities and revision questions. The books provide units of work with the flexibility to allow teachers to personalise subjects to suit their classes. The activity worksheets require science equipment, most of which should be available in the school laboratory, such as student microscopes, Bunsen burners, petri dishes, filter paper and glass slides. The teacher should check for the availability of specific materials needed for each activity. The majority of the worksheets are self-contained.



1	Introduction to forensic science	5
1-1	What is forensic science?	5
1-2	How observant are you?	7
1-3	Looking for differences	9
1-4	Arriving at the crime scene	10
1-5	Removing the evidence	13
1-6	The forensic science laboratory	14
1-7	Going to trial	16
1-8	Microscopes in the forensic laboratory	17
1-9	A crime scenario: Theft at Hilltop Farm	19
1-10	Forensic science word search	21
2	Contact traces	22
2-1	What are contact traces?	22
2-2	Fibres and fabrics	24
2-3	Activity: Looking at fibres	25
2-4	Hair	26
2-5	Activity: Looking at hairs	28
2-6	Using light of different wavelengths	30
2-7	Activity: Examining liquids under UV light	31
2-8	Activity: Looking at flame colours	32
2-9	Chemicals, colours and spectroscopy	33
2-10	Glass	35
2-11	Activity: Glass and refractive index	37
2-12	Paint chips	39
2-13	Activity: Looking at paint flakes	40
2-14	Case study: The Graeme Thorne kidnapping	42
2-15	Contact traces crossword	43
	Answers	45



The word '**forensic**' comes from the Latin word *forensis* meaning '**connected with the forum**'. The forum was the Roman square where the Court of Law and public discussions were held. In some countries, such as Canada and the United States, forensic science is also known as *criminalistics*.

Forensic science is any science that is used to solve crimes, and its results are used in the courts of law. It is based on the belief that whenever a contact is made, there is always some exchange of material. This principle was first suggested by the famous French criminologist **Edmond Locard** who, in 1910, said that a criminal always leaves something at the scene of a crime, or takes something away. In other words, when a person comes in contact with an object or another person, there is always some material transferred from one to the other. This has been referred to as **Locard's Exchange Principle**. These pieces of evidence are for the police to find and for the forensic scientist to analyse.

Things left at a crime scene can include fingerprints, small fibres from clothes, body hair, saliva left on a sealed envelope, or semen in the case of a sexual attack. A suspect can be incriminated if flakes of their skin are found under the nails of a victim who has been attacked. A criminal may take away drops of a victim's blood, some hair or fibres from the victim's clothes or soil from the crime scene on the soles of their shoes. The expression used today for such exchanged materials is '**contact traces**' or 'trace evidence'. Other evidence can include footprints, tyre prints from the get-away car, tool marks from a forced entry, body fluids, ink, paint flakes and oil. In fact anything, no matter how small, that has been altered by the crime can be called a contact trace.



What is forensic science?



Before the 20th century, criminals had it very easy. As long as there were no witnesses, criminals were hardly ever caught. With modern technology today, however, it is almost impossible to commit the perfect crime. The criminal would have to be naked, have a shaved head and body, wear gloves and destroy them and all tools afterwards. Even then, there would be no guarantee that there were no traces left behind or carried away from the scene.

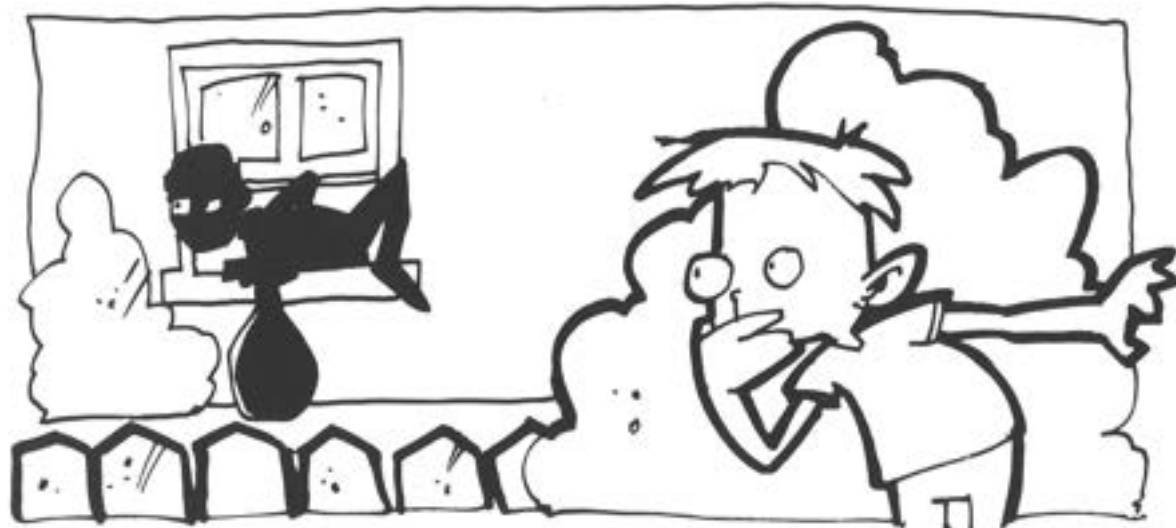
Today, eyewitness accounts are still very helpful in bringing a criminal to justice. A car registration number recorded by an eyewitness can lead to a possible suspect and save the police many hours of work. A good description of a criminal by an eyewitness can lead to the speedy capture of a suspect and eliminate other innocent people from police inquiries.

To be a good eyewitness, a person should be alert and observant, so that they can notice and remember as many details as possible.

?

Questions

1. Explain how forensic science is different to other branches of science.
2. About how long ago do you think that the solving of crimes became a science? Who or what do you think was a major influence on this?
3. Explain why it is more difficult for a criminal to get away with a crime nowadays than it was 100 years ago.
4. How were criminals usually caught before modern forensic science techniques were used to analyse evidence?
5. How can you be a good eyewitness? Give examples.
6. Copy out Locard's Exchange Principle and explain what it means. Give at least 3 examples.
7. Why is a knife used in an assault not a contact trace?
8. Explain why contact trace evidence is so important in solving crimes.





Below is a scene that could occur in any suburban street. Look at it for 30 seconds, then turn over the page and answer the questions in your science book. First write the heading 'How Good a Witness Am I?'





Questions

1. How many vehicles are in the picture?
2. How many children are in the picture? What are they doing?
3. How many people are in the car? Describe them in as much detail as you can.
4. What type of car is it? Is there anything different about the car?
5. What is the man on the other side of the road doing?
6. How would you describe the location of the scene?
7. Describe the parked car.
8. What can you recall about the houses in the street?
9. Are there any animals in the picture? If so, how many and what are they doing?
10. Can you remember anything else about the picture?

Look back at the picture and tick every answer that was correct. How did you go? Would you make a good police witness?

This exercise depends on careful observation and an accurate memory. Unfortunately, when most crimes are committed, witnesses are not expecting anything and are caught unawares. Often, a witness may not even realise a crime is taking place until some time afterwards. However, many offenders have been caught thanks to accurate eyewitness accounts. Good witnesses can lead to possible suspects, or they may help the police investigators look in the right place for possible clues.

11. Imagine you were watching the scene on the previous page. You would be an eye witness.
A few minutes after this scene took place, a police car drove up and told you that they were chasing some people they believe may have taken part in robbing a bank a few minutes ago back along the street. Make a list of at least five separate pieces of information that you saw, that might help them to see if they were involved.
12. How many English words, that are not proper nouns, can you make with the letters in the word 'OBSERVANT'?



To be a good detective or forensic scientist, you must have good observation skills, and be able to detect small differences and objects out of place.

?

Question

Look carefully at the two pictures drawn below. In your science book, under the heading 'Looking for Differences', write down as many differences as you can find. You should be able to find about 15.



Score

- Less than 4 – keep trying
- Over 4 – not bad for a newcomer to forensics
- Over 8 – good detective work
- Over 12 – brilliant detective work



Even the best experts in the world cannot solve a crime without evidence. Therefore, it is essential that the scene of the crime be treated in the correct way.

A. Securing the crime scene

In most cases the first police officer on the scene will be a **uniformed police officer**. They are not usually trained as detectives and so should touch as little as possible of the crime scene and the area in the vicinity. All precautions should be taken to avoid any accidental destruction or contamination of potential evidence.

The officer who arrives on the scene first will ensure the safety of any injured people and of the site itself. This may involve turning off gas, water and electricity supplies. Ideally, the site should be secured with a physical barrier, so that no unauthorised people can enter. This will leave the site as close as possible to the state it was in at the time of the crime, preserving all traces left by the criminals for the officers trained in searching for evidence.

Where it is obvious that a person is dead, the body is not touched or moved in any way until it has been carefully examined and photographed. Where there are injured people involved, the police officer will see to their safety and call an ambulance before asking for the appropriate police backup. While waiting, the uniformed officer may record the names and addresses of any eyewitnesses to the crime.



B. Examining the scene

If a major crime has been committed, then the next person on the scene will probably be the **senior detective** in charge of the investigation. A **police surgeon** and/or a **pathologist** may be called in if a body is involved. Specially trained **crime scene investigators (CSIs)**, also called **scene of crime officers (SOCOs)**, who are experts at looking for evidence, are called in at an early stage of the investigation.

A **police photographer**, or a CSI experienced in photography, will take pictures (still and possibly video) of any victim and the positions of the body. Wounds or other medical features that could help future investigations may also be photographed.

Photographs or videos will be taken to help a court understand the crime. Because the court cannot see the scene as the police investigator saw it, an oral description with photographs and video is the next best option available.



Arriving at the crime scene

1 INTRODUCTION TO FORENSIC SCIENCE



Such a detailed description may even save money and time if it saves the whole court the necessity of visiting the site.

Photographs make a visible record while the evidence is fresh and undisturbed. They should be taken from all angles, as an objective and accurate record of how the scene was left by the criminal. On some occasions photographs have revealed evidence that was not noticed when the scene was first searched.

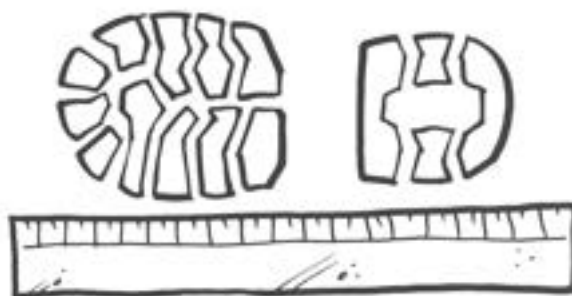
To help record the actual size of an object being photographed, a ruler marked with a suitable scale is usually included in the photo. If photographs of a crime scene are taken from where an eyewitness was standing, they can be most useful later when the witness's statement is evaluated. The photo can confirm if the witness could really have seen what they claim they saw.

A **police draughtsperson** may be asked to sketch plans of the site. On this map, vital information such as the position of doors, windows and furniture; the exact measurements of distances, heights and angles; and the positions of bloodstains, weapons and other evidence is recorded.

A thorough search often takes several hours, and if it includes a large outdoors area, it can even take some days. The search in the Belangalo Forest, NSW, after the first backpacker bodies were found, continued for over a week. In such cases, police or army cadets may be called in and, after an initial briefing about what to look for, they are able to cover a large area very thoroughly.

To avoid contamination, the CSIs put on special overalls and overshoes before starting work. This is to avoid leaving any traces themselves at the scene. They usually try to approach the site using the same route as the first police officer on the scene. On some occasions duckboards are used to approach the scene in order to minimise any interference.

Usually the immediate area surrounding the scene is searched thoroughly first. Things like foot and tyre prints are very perishable and so must be collected as quickly as possible before they are lost due to bad weather or accidental damage.



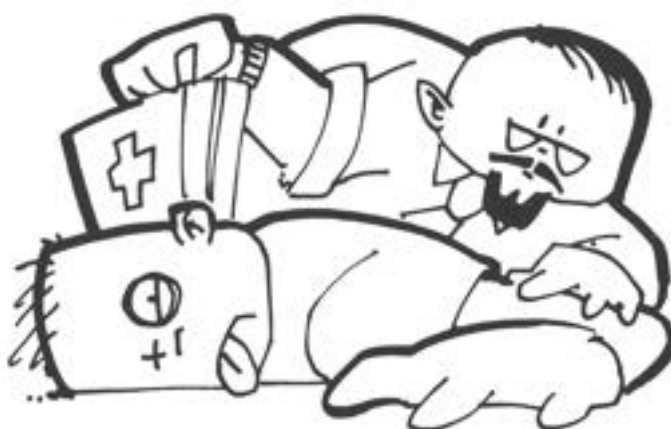
Arriving at the crime scene

1 INTRODUCTION TO FORENSIC SCIENCE



If the crime scene is in a building, the next area to be inspected will be the point of entry. The CSI must try to think as a criminal might, and predict the movements the criminal probably made, so that they will be able to identify the most likely places where evidence may be found and so look there first. For example, they may look for fibres from clothes on a door surround at shoulder height where the criminal may have rubbed their shoulder.

It is the CSI's role to assess the crime scene and prioritise how it will be searched. Often the actual crime scene is searched last. The CSI may call in extra expert help, such as a pathologist to examine a dead body, a forensic biologist to analyse blood splatters or a ballistics expert to look at powder burns or inspect some discarded cartridge cases.



Only after the CSI is satisfied that all possible evidence has been collected will the clean up be able to go ahead. Bodies will be removed to the morgue where a post mortem examination will be made by a police pathologist. All other pieces of evidence are photographed before their positions at the crime scene are recorded on a map. They will then be collected, bagged and labelled before being sent to the laboratory, where they will undergo the appropriate analysis as requested by the officer in charge of the investigation.

?

Questions

1. In the correct order, list four things that should be done by the first police officer who arrives at a major crime scene. Give reasons for your answer.
2. Explain the difference between the roles of the crime scene investigator and the detective in charge of the investigation.
3. Describe at least three procedures that are carried out to prevent contamination of the crime scene.
4. Explain why in most serious crimes a large number of photographs are taken and a map drawn.
5. Why is a small ruler placed in a photograph of a piece of evidence?
6. When would a forensic pathologist be called in, and what is their responsibility?



The evidence from the crime scene must be transported to the forensic laboratory very carefully and systematically so that no evidence is lost, misplaced or damaged in any way. To prevent cross-contamination from the crime scene, investigators wear special overalls and overshoes. Locations associated with a possible suspect are usually searched by different crime scene investigators (CSIs) other than those who searched the crime scene, and any evidence is transported by different vehicles to avoid contamination.

Sometimes in court a defence lawyer may challenge the prosecution, suggesting that an item of evidence tendered in court is not the same as the item that left the crime scene. The lawyer may try to make the members of the jury suspect a possible gap in the **chain of continuity** in the handling of a particular item. The lawyer may raise the possibility of cross-contamination. To prevent a case being lost, all care must be taken to ensure the validity of the evidence.

At the crime scene, all items of possible evidence are labelled and sealed in bags where possible. The label records the location where the item was found, the date and time found and the name of the police officer handing the item over. A laboratory registration number is added at the forensic laboratory, recording all details, including the time of arrival at the laboratory.

This information is entered into a register which has a separate entry for each item, recording all movement of the item and with whom, while in the laboratory. This register can be used in court if the chain of continuity is challenged by the defence lawyers.



EVIDENCE	Description		
	Removed from		
	Address		
	By	Date	Time
	Where		
	CASE		

CRIME SCENE SEARCH EVIDENCE REPORT			
Name of Subject			
Offence			
Date of Incident	Time	AM/PM	
Search Officer			
Evidence Description			
Location			
CHAIN OF POSSESSION			
Received From			
By	Time	AM/PM	
Date	Time	AM/PM	
Received From			
By	Time	AM/PM	
Date	Time	AM/PM	
Received From			
By	Time	AM/PM	
Date	Time	AM/PM	

?

Questions

1. Explain why CSIs wear special overalls and overshoes while at the crime scene.
2. Why do you think it is usual for one CSI to search a suspect's house and another to search the crime scene?
3. Explain why every item of evidence that is removed from the crime scene is enclosed in its own carefully labelled plastic bag.
4. What information is recorded on the outside of the evidence bag?
5. Describe what happens to the evidence once it arrives at the forensic laboratory.
6. Why is it so important to prevent cross-contamination?
7. On the right is a tag that was attached to an item of evidence when it arrived at the forensic laboratory. What important information needs to be added?

BURGLARY at 25 High St, Smithton Found behind the sofa P.C. Jones



The forensic scientist, employed full-time by the police department, is the analyst back in the laboratory. They usually do not attend the crime scene. Their main responsibility is to help the police investigation by using their specialist knowledge and techniques. Forensic scientists usually have a science degree from a university, with specialist training and experience on top of that. They are often not police officers and they do not usually interview witnesses or suspects; interviewing is left to the police officers.

Many forensic scientists specialise in a certain area such as fingerprint identification, DNA profiling or forgery detection. It is the forensic scientist who can identify whether a hair is human or animal and can identify from what part of the body it came. A forensic scientist who specialises in ballistics can often identify the exact gun from which a bullet was fired.

The forensic laboratory houses many pieces of expensive and specialised equipment to help the scientist analyse the evidence.

Once an item arrives at the laboratory, the forensic scientist decides what tests or analysis it should undergo, and in what order. Some tests will destroy part or all of the evidence, while other tests will not affect the sample at all. Obviously, all non-destructive tests should be carried out first. The scientist needs to know the information required by the investigating officers in order to select the appropriate tests.

To prevent cross-contamination in the laboratory, all evidence from the crime scene is stored in a different place to any items taken from a suspect's home or car. Items from the different scenes are analysed by different forensic scientists where possible, and always on different days and in different rooms.





Other experts may be called in to analyse specific evidence or give their opinion on some matter in which they have expertise—perhaps an entomologist employed by CSIRO, a practising dentist or dental technician who has expertise in identifying a person from dental records, or a psychologist who can draw up a profile of a serial killer. Their role in court is exactly the same as a person who is employed full-time as a forensic scientist. Such people are called **expert witnesses**.

?

Questions

1. What would the following experts specialise in? Discuss the types of cases they may be asked to testify about.
 - (a) dentist
 - (b) psychologist
 - (c) entomologist
 - (d) serologist
 - (e) pathologist
2. How may a forensic scientist differ from an expert witness, if at all?
3. How does the role of the crime scene investigator differ from that of the forensic scientist?
4. An unusual red fibre was found caught on the clothes of a victim of an assault. Not long afterwards a suspect wearing a red jumper of a similar colour was brought in for questioning. Some samples of the fibres were removed from the suspect's jumper and taken to the forensic laboratory for testing. It was decided to test the fibres for strength by placing each of them in a clamp and then pulling and measuring at what force the fibre broke. A melting test was also done, to determine at what temperature the fibre would melt. The fibres were examined under the microscope and their colours were compared under ultraviolet light. Discuss the order in which the tests should have been carried out and why they should be done in this order.
5. If the two fibres mentioned in question 4 had the same melting point, strength, thickness and appeared the same under natural and ultraviolet light, what conclusion can you come to regarding the source of the two fibres and the guilt of the suspect?



In court the suspect, the person **police** believe carried out the crime, is put on trial before a jury for major crimes, and before a **magistrate** for less serious crimes. It is often the quality of the police evidence that will influence the opinion of both the jury and the magistrate.

In a jury trial, the **judge** is there to listen and pass sentence if the jury decides that the accused is guilty. The **prosecution lawyer** puts forward the police case and calls any **expert witnesses** to give their opinion in order to convince the jury of the facts of the case.

The **defence lawyer** must try to defend the person charged with the crime by trying to point out any holes in the prosecution's argument and to try to put doubt in the jury members' minds. Only the **jury** can decide if the accused is guilty or not, beyond any reasonable doubt. An expert witness may also be called in by the defence to discredit the case against a suspect.

The scientific evidence must be presented in a way that can be understood by the court, and not in a way that may confuse it. Judges, lawyers and juries do not usually have a deep knowledge of science, so all scientific results have to be presented in non-technical terms. This is why photographs, videos, maps and plaster casts are so important.

Because of the court's lack of scientific expertise, the forensic scientist or expert witness must be prepared to give an expert opinion on what the evidence implies—that is, they must be able to say how likely it is that the crime is linked to the suspect who has been charged. They may give an opinion of the probability of the soil found on the suspect's shoe coming from the crime scene, or the suspect's fingerprint matching one found on some stolen property, hence linking the suspect to the crime.

The forensic scientist not only needs to be qualified and experienced in their field of expertise, they also need to be accepted by the court as having a reliable expert opinion. The court will want to know the extent to which the scientist has specialised in the process under question, and how well the person keeps up to date with developments in their field. They will question any previous experience in similar investigations and the number of times they have been permitted to offer expert evidence at other trials.



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Questions

1. Who decides if the suspect being charged is guilty or innocent?
2. What is the role of the judge in a court of law?
3. Describe the defence lawyer's job and how it differs from that of the prosecution lawyer.
4. What is the role of the expert witness at court?
5. How should scientific evidence be presented in court so that it is easily understood by a non-scientific jury?
6. Describe several techniques used by forensic scientists to show a non-scientific jury exactly what they believe occurred at the scene of a crime.
7. What does a scientist need in order to be accepted as an expert witness at a trial?
8. Who might be called as an expert witness to give their opinion to a court?
9. Why is their opinion necessary?



Still one of the most commonly used and most important tools in the forensic laboratory is the microscope. The simplest microscope is a **magnifying lens** which has been used for over 100 years to examine anything from fingerprints to footprints. It can magnify objects $\times 10$ to $\times 20$.

The common **compound microscope** found in all schools is another useful but relatively inexpensive piece of equipment used regularly in the forensic laboratory. It is used when a magnification of up to $\times 1000$ is needed. It has two lenses, an eyepiece lens at the top through which the scientist looks, and an objective lens at the lower end of the barrel, near the object being examined. Most compound microscopes have three or four objective lenses of different powers of magnification that can be selected by rotating them. Hairs, fibres, glass fragments, pollen grains and fingerprint samples are all commonly examined under such a microscope.

If even greater magnification is required, then an **electron microscope** may be used. Relying on a beam of electrons rather than visible light, this instrument can give images of molecules and even atoms. It can magnify up to $\times 200\,000$ and can be used to examine soil bacteria or pollen grains to show up the smallest difference. Generally these instruments are not found in a forensic laboratory so scientists must send material away to be examined under an electron microscope if they believe that this will reveal important information.

The **stereoscopic microscope** is also a very useful instrument in the forensic laboratory. It has two eyepiece lenses that allow objects to be seen in three dimensions. It has a quite low magnification of about $\times 100$ to $\times 150$, but allows specimens such as soil, hair and fibres that have been collected to be sorted for closer inspection under other microscopes of higher magnification. Quite large objects can be inspected, such as a small paint chip from a car which can be placed vertically in plasticine so that all the layers can be examined.



Fig. 1 Magnifying lens



Fig. 2 A compound microscope



Fig. 3 Stereoscopic microscope

Microscopes in the forensic laboratory

1 INTRODUCTION TO FORENSIC SCIENCE



An unusual microscope that is commonly found in the forensic laboratory, is the **comparison microscope**. It is really two microscopes in one, and is used to compare two different objects at the same time. Its magnification is lower (from $\times 5$ to about $\times 40$) than that possible with the compound microscope. However, it allows the user to adjust the focus of each object separately so that they are viewed through the one eyepiece lens as two halves of one object. Objects most commonly compared in this way are hairs or fibres, fingerprints and marks on bullets. One specimen would be from the crime scene and the other from a suspect or a test firing using a known gun. It was for this latter use that this instrument was invented in the United States in the 1920s.

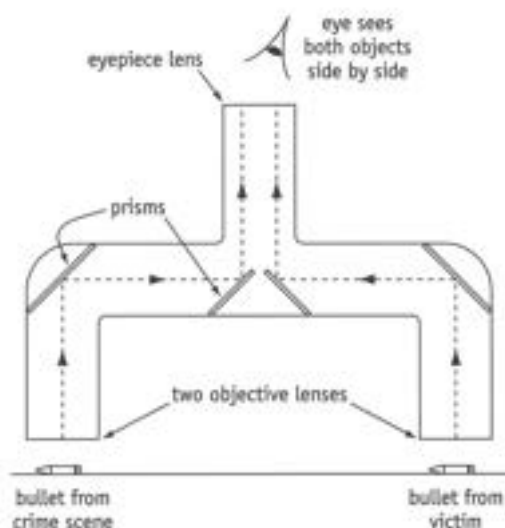


Fig. 4 How the comparison microscope works



Question

Write a summary of the 5 types of microscopes above, mentioning how each is used in solving crimes.



The old green car stopped at the gate while the driver checked the name. 'Hilltop Farm' was painted on the gate, so the driver turned in and drove slowly up the drive. The surface was covered mainly in gravel, but as the car rounded the bend near the house, the driver felt the back wheels slip slightly as they went through a patch of mud on the driveway.

He could see the house now. It was in darkness, so his information was correct. The owners had gone away for a few days, taking their dogs. There was no alarm system either, his informant had said. This was a burglar's dream. He stopped the car at the side of the house, out of sight of the drive, and crept towards the nearest window. It was nicely visible in the moonlight.

He stepped on to the garden to take a look at the window latch. He brushed the flowering wattle tree as he tried to get a better look. He was in luck—there was no lock. He attempted to lever open the window with a large screwdriver he had brought with him, just for this job. Unfortunately, the window was jammed so, using the back of the screwdriver, he gave the glass a sharp tap. It broke easily. He removed the large jagged pieces and climbed through into a lounge room, ripping his jacket as he did so.

As he put his hand on the back of the chair near the window he realised that his hand was bleeding. He must have cut himself on the glass. He wrapped his handkerchief around the wound to stop the bleeding. He would see to it later.

He turned on his torch and was a bit startled to see a pair of cat's eyes staring at him from the chair near the door. Obviously someone would be coming in to feed it. He hoped that it would be much later, but better not make too much mess and hopefully his crime would not be noticed until the owners returned in a couple of days. At least cats don't bark.

He moved towards the desk and started searching through the drawers. He found a few coins, a medal and a small brooch which he put in his pocket. His eyes lit up when he saw the cheque book. There was one cash cheque, made out for only \$5. The signature was there too, but it was not an easy one to copy. However, there would be enough room between the two words to add the word 'hundred', and to add two zeros after the number five in the figures. He would try to cash it before the owners came back from holidays.

Suddenly, he thought he heard a car. He quickly turned off his torch and could see car headlights coming up the drive. He could feel the sweat on his hands and the hair standing up on the back of his neck.

He moved across to the window, steadying himself on the table as he went. He tripped over the cat and fell into the chair where it had been

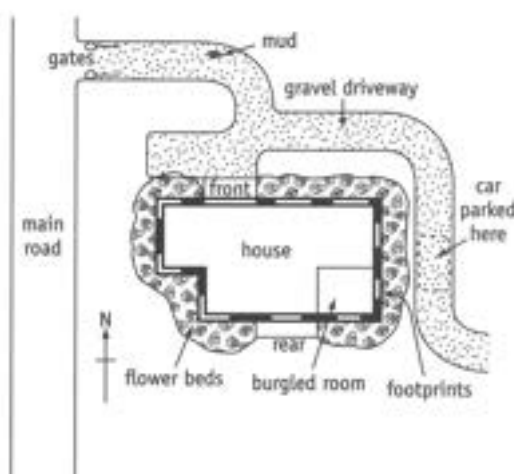


Fig. 5

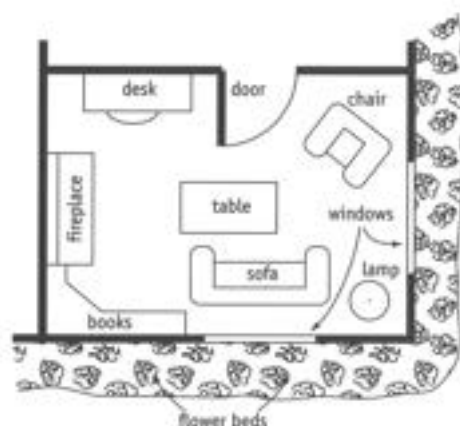


Fig. 6

A crime scenario: Theft at Hilltop Farm

1 INTRODUCTION TO FORENSIC SCIENCE



sitting. He stood up quickly and climbed out of the window. He reached his car just as a key turned in the lock of the front door.

Without turning on his lights, the thief was able to roll down the drive for a little way before turning on his engine and then driving away, still in the dark. The moonlight was a great help but it disappeared behind a cloud just as he was near the gate. He felt a bump and heard a crunching noise as his car scraped against the gate.

As he turned on to the road he turned his headlights on. They still worked, but he supposed he would be left with a dent and possibly a broken indicator light. As he drove off, the oil warning light started to glow on the dashboard. His engine had been losing oil lately, and he had intended to top up the oil earlier. He was relieved to see that he was not being followed. Maybe he had got away with it.

The next day, after changing the cheque to \$500, he headed for the farmer's bank. He felt nervous as he approached the teller though he was sure that the family wouldn't have missed the cheque yet. However, when a policeman entered the bank he panicked, and ran off.

Without looking back, he walked quickly around the corner and entered the next shop which he knew had a back entrance to the lane and car park. As he passed through the shop, he took off the navy parka that he had been wearing, and dropped it in the rubbish skip that was parked beside the back door. He now wore a pale blue windcheater that he was sure would not link him to the bank or the crime at Hilltop Farm. He felt quite safe now.



?

Questions

1. Do you think that the thief has got away with it? Why do you think this?
2. In your science book, under the heading 'Theft at Hilltop Farm', make a list of all the clues the thief has left behind. There are at least 10. Next to each item write what you think the crime scene investigator will do with all these clues. If you have been provided with a map of the crime scene, mark all the clues on the map.
3. Can you think of any contact traces that the thief may have unwittingly carried away from the crime scene which could link him with the crime? Make a list of all the things you can think of. Explain how the police could use them as evidence.
4. Describe any other evidence, other than trace evidence, that may link the thief to the crime scene.
5. Once the police bring the thief in to the police station to question him as a possible suspect, what questions do you think they would ask him?
6. Which of the suspect's belongings might the police ask to inspect? Why do you think this?
7. What type of police officer do you think may have been the first called to the scene and what do you think this person would do there?
8. Which specialist police officers might have been called to the scene? What would be their task?
9. Describe some of the things the thief may say when being questioned, to try to explain away some of the evidence and so maintain his innocence.
10. Which evidence do you think the thief will find hardest to explain? Give reasons for your answer.



Look carefully at the grid and find the words listed below. If these words are shaded in, there should be 24 letters remaining. These letters spell out a sentence that is the basic principle of forensic science.



- | | | | |
|---------------|------------------|--------------------|-----------------|
| 1. analyst | 8. forensic | 15. crime scene | 21. murder |
| 2. ballistics | 9. forgeries | 16. criminalistics | 22. oil |
| 3. blood | 10. glass | 17. Edmond Locard | 23. pathologist |
| 4. breaking | 11. hairs | 18. evidence | 24. theft |
| 5. bullet | 12. impression | 19. expertise | 25. toxicology |
| 6. cast | 13. investigator | 20. fingerprints | 26. witness |
| 7. court | 14. microscope | | |

Basic principle of forensic science:



In 1910, the French criminologist Edmond Locard proposed his '**Exchange Principle**'. He stated that 'every contact leaves a trace'. The things he was talking about include fingerprints, hairs, fibres, glass and paint fragments, even pollens.

We constantly drop hair from our bodies and shed fibres from our clothes. If there is a cat or a dog in our home, its fur can be transferred on to our clothes and carried to a new location. Pollens from plants around our home or workplace can land on our clothes and be carried away to be deposited at another site. A burglar who breaks a window to enter a building is often showered with minute fragments of glass which can get caught in the seams and pockets of clothing. All these very small items of evidence are referred to as **contact traces** or **trace evidence**, and will be meticulously searched for by the crime scene investigators (CSIs). It is almost impossible for a criminal and a victim to be in contact and not exchange particles of some sort. This 'exchange principle' is the basis of all modern forensic science, with the particles that are exchanged being the key to solving the crime.

Searching for trace evidence

At a crime scene, after the large, easily seen objects like weapons and clothes have been removed, the area may be vacuumed so that all loose threads, hair and other small trace particles will be caught in a special container called a **vacuum trap**. Before moving a dead body to the mortuary where the post mortem will be carried out, the forensic pathologist wraps the head, hands and feet in separate plastic bags, so that no small traces may drop off and be lost before a thorough inspection can be made. Any debris under the fingernails from a possible struggle could also yield important evidence, as could any foreign particles that have been breathed into the air passages.

The CSIs may press a type of sticky tape all over the victim's clothes, the carpets, curtains and sofas to collect any loose fibres or hairs. Sometimes, clothes are shaken or beaten above a large funnel in the forensic laboratory examination room so that any loose fibres, hairs, pollens, glass or other particles that may have become attached to the garments can be collected. They are then bagged and labelled before being sent to the forensic laboratory to be examined under a stereoscopic microscope. Any particles that are present in particularly high numbers are given very close attention. Why would this be so?

The **stereoscopic microscope** is a very versatile and frequently used instrument. Because its focus is very flexible, it is extremely useful for sorting through debris from a crime scene to locate possible contact traces. The big distance between the stage holding the specimens and the objective lens means that it can be used for quite large objects. Its magnifying power ranges from $\times 100$ to $\times 50$, which is quite low compared to the conventional compound microscope, which ranges from $\times 4$ up to $\times 1000$, but this low magnification is quite sufficient for sorting.

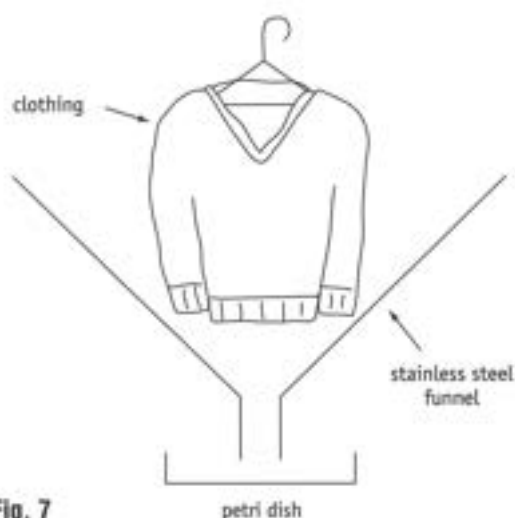


Fig. 7

What are contact traces?

2 CONTACT TRACES



?

Questions

1. List four examples of a contact trace. Explain how each could be used in helping to solve a crime.
2. Describe three ways in which contact traces can be collected from the crime scene, and explain how they are treated once they are collected.
3. Explain why the stereoscopic microscope is the most suitable instrument for sorting through fine debris from a crime scene.
4. How is a dead body prepared for transport from the crime scene? Why is it done this way?
5. Where would a thorough inspection of a body be made and by whom is it carried out? Describe the type of trace evidence that person is looking for.
- 6.



Across

4. Who proposed the 'Exchange Principle'?
7. Term used for microscopic evidence.
10. Trace evidence shed from clothes.
12. Container that collects vacuumed trace particles.
13. Size of trace evidence.
14. Part of clothes that can hide glass traces.



Down

1. Every contact leaves a
2. Trace evidence when window is broken.
3. Place where crime occurs.
5. Where dead bodies are examined.
6. Trace evidence shed by human body.
8. Under finger- is examined.
9. This is vacuumed for evidence.



If loose fibres found at the crime scene have not come from the victim, they may have come from a large number of sources—from the perpetrator's clothes, from carpet or curtains where the crime occurred, or even from a rug or blanket in which the body was wrapped for transportation, if the body was moved. Also, fibres from the victim may have been deposited on the perpetrator, and these will be looked for on all suspects. Any such fibres found could connect the suspect with the crime scene.

Fibres can be either **synthetic** (man-made)—such as nylon, rayon, terylene, polyester—or they can be **natural**, being made from either plants (cotton and linen) or from animals (wool, silk and mohair). Generally, under the microscope, natural fibres have a rougher surface, taper from root to tip and appear somewhat irregular. Cotton fibres are actually collapsed tubes, while wool fibres, like hair, have scales. On the other hand, synthetic fibres are smooth and even and have a much longer length (staple).

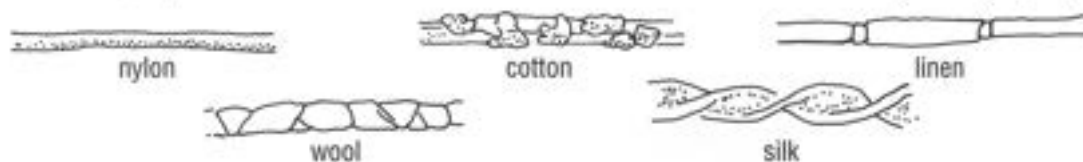


Fig. 8

Fibres of the same type may vary in thickness, depending on how they are made. Their strengths can also vary. Often, both the strength and thickness of any fibres found at a crime scene are measured in the forensic laboratory and compared with any found on a suspect. The melting point of a synthetic fibre is usually measured and its colour analysed. If two fibres look the same under the microscope, have the same thickness, strength, colour and melting point, then they probably both came from the same source. If the fibres are uncommon, then two matching fibres are even more likely to have come from the same garment.

Sometimes, CSIs are fortunate enough to find a piece of fabric instead of only a few fibres. Knitting and weaving can form a number of different patterns. One crime was solved when the weave pattern of a garment was found imprinted on soft putty around a broken window. In another case, a driver involved in a hit-and-run accident was caught when the weave of the victim's clothes left its pattern on the greasy surface of the car's bumper bar.

The distinctive shape of a piece of fabric found at a crime scene may be compared to the suspect's clothes. If a hole of the same shape and size is found, it will show that the fabric fragment came from a particular garment.

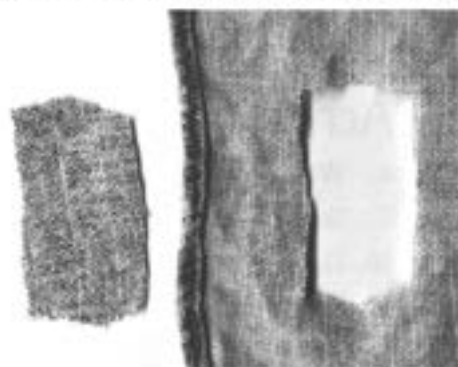


Fig. 9 piece fits in hole exactly

?

Questions

1. Explain how fibres can be important evidence in the solving of a crime.
2. Make a sketch of the fibres above. Describe the differences between natural and synthetic fibres.
3. Using the library, try to find the names of at least 12 different fibres: four obtained from animals, four from plants and four synthetic. If possible, make a sketch of their appearance. Also, try to find out how one of the synthetic fibres is made, and describe the process.
4. Describe five different tests for comparing two fibres that may be carried out in the forensic laboratory. Explain the order in which these tests would be carried out, and why they are done this way.
5. Explain at least two ways in which a small piece of fabric found at a crime scene might be useful in solving a crime.



In this activity you will examine the appearance of some different types of fibres on labelled slides under the microscope. You will then use these to identify two unknown fibres.



Materials

- normal student compound microscope
- prepared and labelled slides of a number of different fibres
- two slides of some unknown fibres.



Procedure

In your science book, under the heading 'Looking at Fibres', rule up the following table, with a separate box for each slide you examine. Inspect the named slides and make a simple sketch of each type in the table.

NATURAL FIBRES				
	wool	cotton	linen	silk
SYNTHETIC FIBRES				
	rayon	polyester	nylon	terylene

Now look at the two unknown fibres and, using the sketches of the named fibres, identify the type of each fibre, and say whether you think it is natural or synthetic.

UNKNOWN FIBRES		
		



Questions

1. Describe the main differences in the appearance of synthetic fibres and natural fibres.
2. What type of fibre do you think the unknown fibres were? Explain what led you to make this identification.
3. What are three other tests that could be carried out to confirm your identification?
4. Describe a crime scene where fibres would be important forensic evidence.
5. Name five fibres that are not mentioned in this activity. For each one state whether it is a natural or synthetic fibre.



Although hair cannot usually identify a single individual as fingerprints can, hair found at a crime scene can be very useful. Besides giving some indication of age, race and sex, hairs found at a crime scene can strengthen the links between a suspect and that scene.

Human hair can be classified into six groups, according to its origin:

- **Hair from the scalp** tends to have a circular cross-section, often with somewhat split ends.
- **Beard hair** often has a more triangular cross-section.
- **Eyebrows and eyelashes** also have a circular cross-section, but their ends are tapered.
- **Under-arm hair** has a more oval cross-section than the other types.
- **Pubic hair** is often short, curly and stiff, with an oval or triangular cross-section and short roots.
- **General body hair** is finer and softer than pubic hair, but is also curly and either triangular or oval in cross-section.

Hair is composed mainly of the protein **keratin**, which is very difficult to destroy except by fire and certain chemicals. It will remain preserved in the grave longer than most of the body except for the bones. Microscopic inspection of hair can also indicate whether hair was pulled out, cut off or whether it just fell out naturally.

Every person's hair is also unique in the trace elements it contains. These chemicals reflect the food you eat, your work and home environment, any drugs and poisons you may have consumed as well as the shampoos and other chemicals like dyes and bleaches that you may have put on your hair. In the 1960s, when some of Napoleon's hair was obtained and analysed more than 100 years after his death, it was found to contain many more times the normal amount of arsenic. In his day arsenic was thought to improve the complexion, but it was also a favourite method of poisoning as its taste was difficult to detect.

A strand of hair consists of three layers:

1. the central core, called the medulla;
2. the coloured layer, called the cortex, surrounding it; and
3. the outer layer of overlapping scales, called the cuticle.

Examination of all three layers can help the forensic scientist to identify the hair. The pattern of overlapping scales in the cuticle is very useful in identifying the type of animal it came from, and the medulla, which can vary markedly in different people and animals, can be classified into three groups (see the diagram below).



Fig. 10

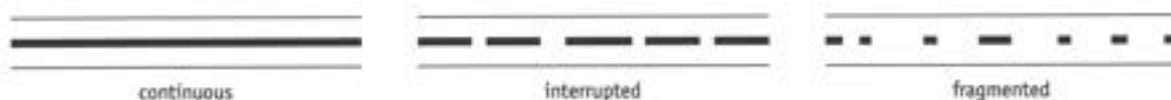


Fig. 11

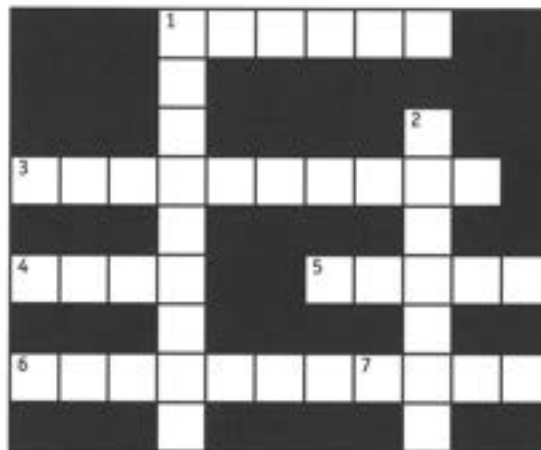


In some humans the medulla may be absent, but the most common is a fragmented medulla. Humans of the Mongoloid race tend to have a continuous medulla. It is usually cylindrical in shape in humans, but this is not true for many animals. For example, the medulla in cat hair is like a string of pearls.

?

Questions

1. Write a paragraph explaining all the information that might be gained about a person whose hair was found at a crime scene.
2. What are the six parts of the human body that have hair with different characteristics?
3. Describe two types of information that can be found from a hair sample.
4. Of what substance is hair made? How durable is this substance?
5. Name the three layers of a strand of hair, and describe where each is found.
6. What information can a forensic scientist find by inspecting the medulla of a hair sample?
7. How can inspection of the cuticle help in identifying hair?
- 8.



Across

1. Coloured layer of hair.
3. Most common type of medulla in humans.
4. Cross section shape of under arm hair.
5. Type of human hair with triangular cross section.
6. Name of outer layer of hair.
7. One of the few ways hair can be destroyed.



Down

1. A race that has a continuous medulla.
2. The type of protein that hair is made of.



In this activity you will inspect a number of labelled slides of different hairs under the microscope then, using the labelled slides, try to identify some unknown hairs.



Materials

- normal compound microscope
- labelled, prepared slides of some different hairs from different parts of the human body as well as some common animal hairs
- some slides of unknown hairs.



Procedure

1. In your science book, under the heading 'Looking at Hairs', draw up the following table.

Type of hair	Description
Unknown 1	
Unknown 2	

2. Look at each hair under the microscope, then write down a description of its appearance in the right-hand column of the table.
3. Using the information you gained from examining the known hairs, try to identify the unknown hairs. Were they human or animal? If human, what part of the body? If animal, what type of animal?



Questions

1. What differences did you notice in the hairs from different parts of the body?
2. How did the animal hairs differ from the human hair?
3. Where do you think the unknown hairs came from? How difficult was it to identify them?
4. What features on the hairs enabled you to tell where each had come from?
5. (a) Look at the diagrams below of the medulla of three different animals and that of a human, magnified $\times 450$. Try to identify where the hair found at the crime scene came from.

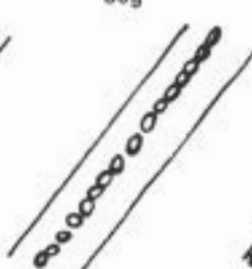
human head hair



cat



dog



mouse



crime scene



Fig. 12

Activity: Looking at hairs

2 CONTACT TRACES



- (b) Try to identify the hair from the crime scene using the diagrams below showing the overlapping scales of the cuticle.

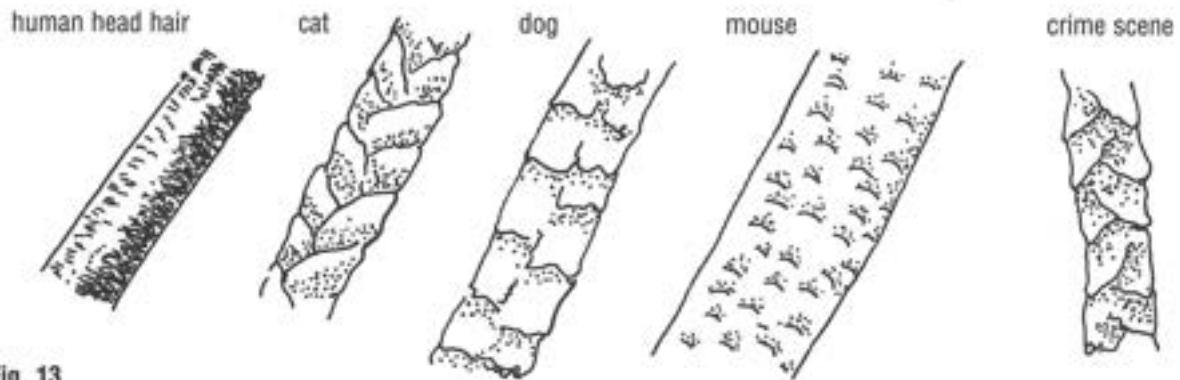


Fig. 13

6. Looking at Hair

Find the words listed below in the grid. The words may be found forwards and backwards, up or down, and even on a diagonal. If you find all words correctly, then the 30 letters left should spell out a sentence:

H	A	I	R	F	W	O	R	B	E	Y	E	L
N	I	T	A	R	E	K	H	A	I	X	M	A
R	S	D	R	A	E	B	A	R	E	E	I	Y
S	M	O	N	G	O	L	O	I	D	T	N	E
L	M	P	O	M	S	R	P	U	Y	R	I	R
A	A	T	A	E	N	C	L	T	E	O	E	E
C	C	V	F	N	O	L	A	N	S	C	T	L
I	T	I	O	T	A	A	C	L	O	C	O	C
M	R	T	U	E	T	R	S	R	E	A	R	I
E	C	C	E	D	B	L	E	A	C	H	P	T
H	I	N	T	E	R	R	U	P	T	E	D	U
C	S	U	O	U	N	I	T	N	O	C	S	C

- | | | | |
|---------------|-------------|-----------------|---------------|
| 1. beard | 7. cut | 12. fragmented | 17. Mongoloid |
| 2. bleach | 8. cuticle | 13. interrupted | 18. oval |
| 3. chemicals | 9. dyes | 14. keratin | 19. protein |
| 4. continuous | 10. eyebrow | 15. layer | 20. scalp |
| 5. core | 11. fire | 16. medulla | 21. hair |
| 6. cortex | | | 22. scale |



When sunlight passes through a prism, it separates into a whole series of different colours that we call the visible spectrum. These are the same as the colours of a rainbow. This was discovered by Sir Isaac Newton in 1666. Each colour is associated with a different wavelength.

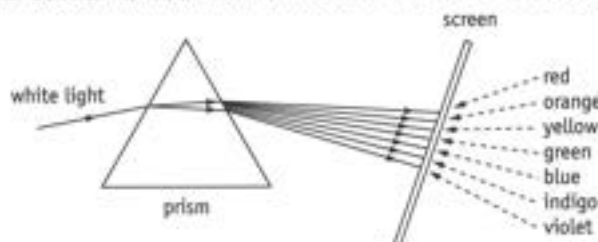


Fig. 14

The visible spectrum is only a small part of the **electromagnetic spectrum**. On both sides of these visible colours are other rays that are not visible. The region beyond the red colour, with wavelengths longer than 10^{-6} m, is the infra-red region (see figure 15). Infra-red radiation is used in heat lamps and high speed ovens. Waves that have even longer wavelengths than the infra-red rays include microwaves and radio and television waves.

The region beyond the violet colour, with wavelengths less than 10^{-7} m, is the ultraviolet region. These rays have a lot of energy and can hurt your eyes badly if you look at them. There are ultraviolet rays in sunlight and they tan the skin; however, too much can cause sunburn. Radiation with even shorter wavelengths than ultraviolet light includes X-rays and gamma rays used in medicine.

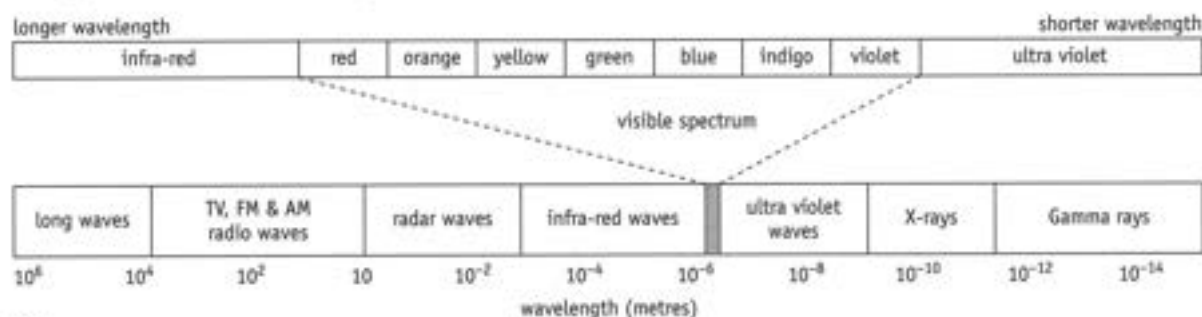


Fig. 15

The forensic scientist uses both the infra-red (IR) and ultraviolet (UV) parts of the spectrum as well as visible light. The ultraviolet light causes some substances to **fluoresce** or glow when it is shone on to these items. We say such substances or materials are **fluorescent**. Some objects continue to glow after the light has been removed. We say these things are **phosphorescent**.

Substances like some washing powders contain small amounts of fluorescent chemicals so that they glow a bluish-white in sunlight, giving the impression of whiteness. Some oils also fluoresce under UV light, so oils that may appear the same under normal sunlight can look quite different under UV light.

Other contact traces that forensic scientists may examine and compare under UV light include fabrics and fibres, paint samples, inks and dyes, and even glass fragments.

?

Questions

1. Explain what the visible spectrum is, and how it can be displayed.
2. What are the colours of the visible spectrum, in order from longest to shortest wavelength?
3. Describe how ultraviolet and infra-red are different to visible light.
4. Explain how ultraviolet radiation can be used in the forensic laboratory to help solve a crime.
5. What is the difference between fluorescent and phosphorescent?



In this activity you will examine a collection of oils, such as gear box, baby, cooking, paraffin and sewing machine oils. Other liquids that could be inspected include petrol samples, tonic water, washing detergent and other household substances. These should be examined under natural and UV light.

Note: Take care to **never** look directly at a UV lamp. Also take care **not** to have a **naked flame** near **flammable liquids** such as oil and petrol.



Materials

- selection of different oils and similar liquids
- some absorbent paper such as filter paper
- UV lamp.



Procedure

1. Draw up the following table in your science book, under the heading 'Examining oils and other liquids under UV light'.

Type of oil or liquid	Appearance in natural light	Appearance under UV light

2. Place a few drops of each of the liquids on the absorbent paper or inspect each liquid in its clear container. Examine each substance, first in natural light, then under the UV light.



3. Record the appearance of each substance under each light in your table.



Questions

1. Which substances showed the greatest change in appearance when viewed under UV light?
2. List all the substances that showed a moderate change in appearance.
3. Which substances were non-fluorescent?
4. Describe a possible crime in which a forensic scientist may examine evidence under UV light.



In this activity you will investigate the flame colour of a number of metal compounds, including several containing the same metal ion component and several having the same non-metal ion component.

Materials

- a number of labelled vials containing some different metal compounds (salts)
- two vials of unknown salts
- Bunsen burner
- spectroscope
- platinum wire (one per compound)

Note: Because it is difficult to remove the compounds from the platinum wire, it is recommended that one wire remain with each salt throughout the experiment, and that the students move around the room, testing each compound at its own bench. The cleaning can then be done without students handling concentrated hydrochloric acid.

Procedure

1. Rule up the following table in your science book under the heading 'Looking at Flame Colours'.

Name of salt	Flame colour	Observations

2. Collect all apparatus and light the Bunsen burner.
3. Wet the wire with a little water then dip it into the salt. Hold the wire in the flame and look for the first colour observed. Repeat if unsure.
4. Repeat step three above with all the other salts. Then, use the information from the named salts tested to identify parts of the compound in the two unknown samples.



Fig. 16

Questions

1. Many compounds had different flame colours. Do you think the distinctive colour is due to the metal or non-metal ions (parts) of the compound? Explain what has led you to this conclusion.
2. Was it possible to identify the other ion (part) of the compound? Explain your answer.
3. Suggest a situation in which a forensic scientist could use flame colours to help solve a crime.
4. Using chemistry textbooks from the library, find out what a salt is, then an ion, and what causes the distinctive flame colour in some salts. Make a list of as many different metals as you can, each with its own unique colour.
5. Why was the wire made wet in step three?



Fig. 17

A simple line drawing of a girl's head in profile, facing right. She has a ponytail with a bow. She is holding a flashlight up to her right eye, with the beam of light shining towards the viewer.

bright red shades pale mauve shades

Instant Lessons in Forensic Science Book 1

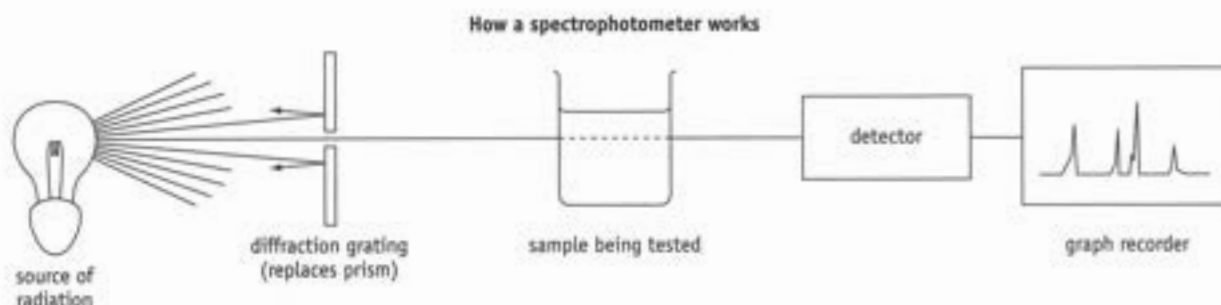


Fig. 19

In the forensic laboratory, scientists use more sophisticated instruments such as the **spectrophotometer**, which is based on the same principle as the spectroscope, but can actually measure the intensity of each wavelength in the sample, and so can identify the proportion of each element in a substance. This is very useful when comparing two substances that appear very similar and have to be chemically analysed.

The advantage of this technique is that only small amounts of the substances are needed, but the disadvantage is that the substance must be destroyed in order to find what chemicals are in it.

?

Questions

1. Colour in the colours of the continuous spectrum in Fig. 18, fading one colour into the next. Use coloured lines to colour in the line spectra of the three metal elements.
2. Explain what a spectroscope is, and what it can be used for.
3. Describe how the line spectrum of sodium is different to that of potassium.
4. Why would a forensic scientist prefer to use a spectrophotometer rather than a spectroscope?
5. Explain under what circumstances a forensic scientist may need to identify an unknown salt.
6. What is one advantage of using the spectrophotometer in forensic science, and what is one disadvantage of this technique?



Glass is a material that is made mostly from sand. For practical uses it is usually clear and transparent. It can be made to be heat resistant, coloured, stronger, more brittle or more flexible by just changing the amounts of the other chemicals that are added to the silica from the sand. The mixture is heated to a very high temperature so that all the substances melt together. When forensic scientists are asked to compare fragments of glass from a crime scene, there are a number of tests which they may carry out. These include **inspection under UV light**, and a measurement of its **density** and **refractive index**.

Some glass samples will fluoresce under UV light, so this test would be carried out first. Density is a measure of how closely the particles of a material are packed together. If a type of glass contains a lot of lead then it would tend to be denser than one with very little lead. The **density** of a sample can be compared by finding in what liquids a sample will float. If the glass is placed in a liquid like water or methylated spirits and sinks, we know it is denser than these liquids. If it floats on top of a dense liquid like glycerine, then we know it is less dense than glycerine. If the glass neither sinks nor floats in a certain liquid, or mixture of liquids, then its density must be exactly the same as the liquid. Now, if two glass fragments were both suspended in the same liquid, then they would have exactly the same density, and may have come from the same source. Comparison of densities is not used as often in the forensic laboratory as in previous years, because glass manufacture today is more automated and density characteristics do not vary as much as they did years ago when each factory had its own 'recipe'.

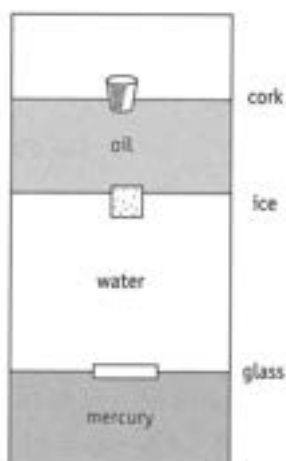


Fig. 20

Refractive index is a measure of the amount that light is refracted or bent as it passes out of air and into another transparent substance such as glass or water. This bending of light occurs because its speed is slowed down. The speed of light in air is about 300 000 km per second, while in water it is only about 225 000 km per second. In glass it is even slower at approximately 200 000 km per second.

The refractive index is actually the ratio of the speed of light through a vacuum to its speed through the material. The refractive index of a vacuum is exactly 1.0, while the refractive index of most glass is approximately 1.5. To determine if two pieces of glass have come from the same source, their refractive indices can be compared.

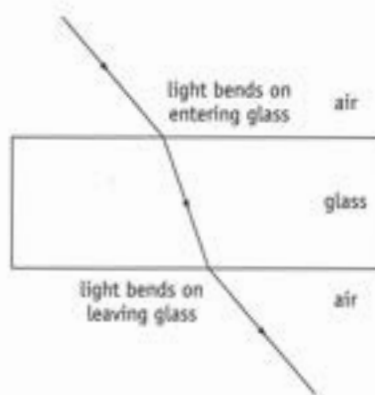


Fig. 21

If a piece of glass is put into a dish of water, the outline of the glass can be easily seen because the water and glass have different refractive indices. It is found that some silicone oils have a refractive index slightly higher than that of glass. If the oil is heated, then its refractive index will gradually decrease, and become the same as the glass. At that temperature, any glass that is sitting in the liquid will no longer be visible as the refractive index of the heated oil is the same as that of the glass. If the oil is heated further, then the glass should be visible again as the refractive index of the oil becomes lower than that of the glass.

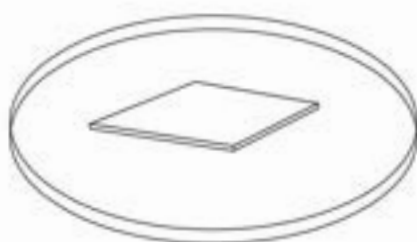


Fig. 22

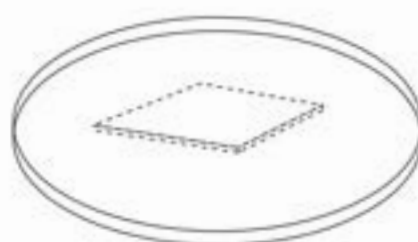


Fig. 23

?

Questions

1. Explain how glass is made. What different properties can it have, and how is this done?
2. Describe three types of tests that are likely to be carried out if you were comparing glass fragments found at a crime scene with others found on a suspect.
3. What is density? What would make some glass samples denser than others?
4. Explain how the density of two fragments of glass can be compared.
5. What is the refractive index of a transparent substance?
6. Describe how the refractive index of two fragments of glass can be compared using a silicon oil whose refractive index is just over 1.5. Explain why this method works.
7. The piece of glass on the right was found at a crime scene. The three cars A, B and C were brought in by the police as possibly having been involved in the hit and run accident being investigated. Which broken headlight might the piece from the crime scene come from?

Missing parts of headlight are shaded



Fig. 24



In this activity you will try to solve a crime that occurred recently when elderly Mr Ino Scent was attacked and his wallet stolen. In the scuffle his prescription spectacles were broken. A suspect, Miss Vera Lightfingers, was brought in for questioning. Her clothes were searched and a large number of minute glass fragments were found in the seams of her jeans. She explained that these were probably from the glass that she had broken that morning. You have been given the evidence which includes the glass fragments found on Miss Lightfingers's clothes (Exhibit A), glass from Mr Scent's spectacles (Exhibit B), and some glass fragments found in Miss Lightfingers's rubbish bin (Exhibit C). Your task is to compare the refractive index of the glass samples, and determine if the glass found on the suspect's clothes matches either of the other two samples.



Materials

- petri dish of silicone oil
- tweezers
- three different types of glass samples:
 - (i) glass from suspect's clothes (Exhibit A)
 - (ii) glass from MR Scent's spectacles (Exhibit B)
 - (iii) glass from Miss Lightfinger's rubbish bin (Exhibit C)
- a hot plate



Procedure

1. Rule up the following table in your science book.

Glass sample	When glass disappeared	When glass reappeared

2. Using the tweezers, place the three different glass samples in the oil.



Fig. 25

Activity: Glass and refractive index

2 CONTACT TRACES



- Place the petri dish on the hot plate (remember which sample is which).

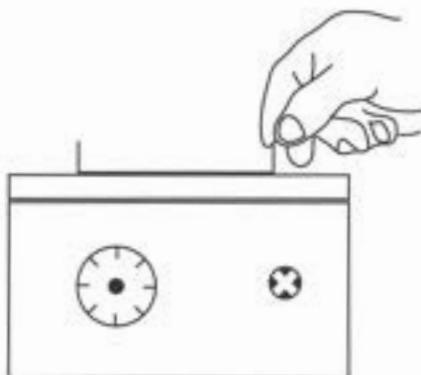


Fig. 26

- Gently warm the oil, continuously noting the appearance of the glass fragments.
- Record the time at which any of the glass fragments seem to disappear or reappear.

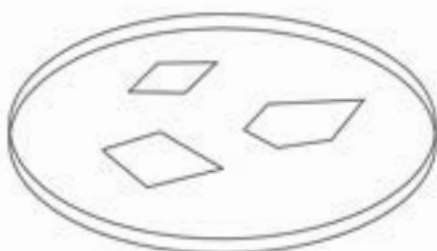


Fig. 27

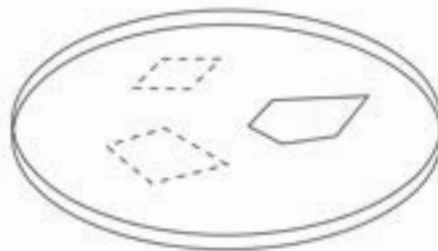


Fig. 28



Questions

- Did any glass samples disappear and reappear at the same time? If so, which ones?
- What does this indicate about the glass found on Miss Lightfingers's jeans?
- As the forensic scientist, what would you report back to the police who were investigating this crime? Is the suspect guilty or innocent?
- Describe any other tests that could be done on the glass to help solve this case.
- How many English words that are not proper nouns, can you make from the letters in the words 'REFRACTIVE INDEX'?



Small fragments of paint can be chipped off when a hard object strikes a painted surface. For example, this may occur when a car strikes something or someone. If a thief tries to jemmy open a window, the tool may remove some flakes of paint. These flakes can be unique if the building is an older one that has been repainted several times.

If such paint chips are found, they can be placed edge-on under a stereoscopic microscope, and the different layers of paint can be examined. It is not unusual for some older cars to have up to ten layers of paint, including binding materials. In each case, the colours and the order of these colours are unique for different makes and models of cars. The scientist is able to examine the flakes for colour, texture and the sequence and thickness of the colour layers.



Fig. 29 Paint flake in plasticine being viewed in a stereoscopic microscope.

These days, the colour layers for all models of cars are supplied by the car manufacturer, and are recorded on a computer database. This allows the forensic scientist to quickly discover the colour, make, model and year of manufacture of the car from a paint chip.

If the paint chips found at a crime scene match the paint on a suspect's car, then that car was probably at the scene. Often dents or damage are further evidence that incriminates the suspect.

Some paints and binders used on the car body can be dissolved in some solvents such as acetone (found in nail polish remover) or butane (cigarette lighter fuel). Paint chips can be placed in a number of different solvents and examined to see if there is any seepage of colour from any of the layers. If the same layer in two different chips dissolves in the same solvent, and the order of the other layers is the same, it can be concluded that the two chips are probably from the same object.

?

Questions

- Describe two tests that can be carried out to determine if two paint chips are from the same source.
- If the layers of paint, fillers and binders of a paint flake taken from a crime scene exactly match the order and thickness of the layers in another paint flake from a suspect's car, what conclusion can an investigator make regarding the suspect?
- What other information would be needed before an arrest should be made?
- If two paint flakes both had the same eight layers of the same colours, in the same order, but the thickness of some of the layers differed, could you conclude that they probably came from the same object? Explain the reasons for your answer.
- The paint flakes (a) to (d) below were also found at the crime scene. The paint flakes A to D were removed from four suspect cars.

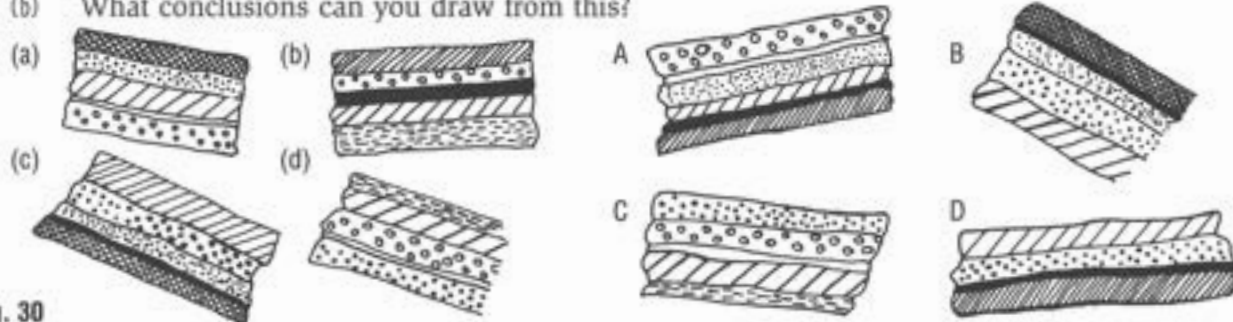


Fig. 30



In this activity you will be given a number of paint flakes collected from several damaged suspect vehicles, as well as one that was found imbedded in the wooden gate-post at the site of a house burglary. The CSI thinks that the gate might have been hit by the get-away car.

Your task is to see if the chip from the crime scene matches any of the suspect cars by examining the paint layers and testing the solubility of the different layers.



Materials

- a number of paint flakes labelled to indicate their source
- a microscope (preferably stereo, but a compound one will do)
- small amount of plasticine
- microscope slide
- ceramic tile with indentations
- acetone
- ethanol
- butane.

Note: Acetone, butane and ethanol are all flammable, so only a small amount should be used in the little indentations in the tile and they should be kept away from flames.



Procedure

1. Place the paint chip, edge-on, in the plasticine on the slide.



2. Adjust the microscope to low power and examine the paint chip layers.

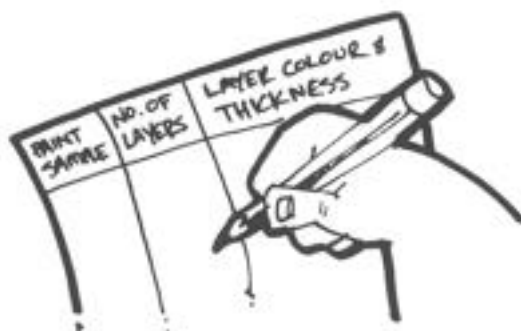


Activity: Looking at paint flakes

2 CONTACT TRACES



- Record the numbers of layers of paint, their colours and thickness in a table in your science book.
- Place a small part of the paint chip in 3 of the indentations of the spotting tile. Add a drop of acetone to one, ethanol to the next one and butane to the last one. Look for any sign of colour bleeding or dissolving. Repeat for all the paint chips and record your results.



?

Questions

- Draw up a table in your science book showing how all the flakes compared in each of the tests.
- Was one of the suspect chips identical to the one found at the crime scene? If so, which one?
- List all the similarities between the two.
- Would this evidence be enough to incriminate the driver of this car? Discuss the reasons for your answer.
- What other information would be useful to the police investigating this crime?
- Wordsearch

E	P	O	C	S	O	R	C	I	M	O	E	R	E	T	S
N	P	I	H	C	U	O	F	R	A	G	M	E	N	T	S
O	R	E	Y	A	L	S	T	H	I	C	K	N	E	S	S
T	O	O	L	O	C	N	P	S	O	L	V	E	N	T	K
E	C	X	U	A	E	E	N	E	C	S	E	M	I	R	C
C	Z	R	R	D	S	P	I	H	C	T	N	I	A	P	O
A	B	I	N	D	I	N	G	M	A	T	E	R	I	A	L

- | | | | |
|---------------------|----------------|-----------------|----------------------|
| 1. acetone | 5. colour | 9. layer | 13. stereomicroscope |
| 2. binding material | 6. crime scene | 10. lock | 14. suspect |
| 3. car | 7. dent | 11. paint chips | 15. thickness |
| 4. chip | 8. fragments | 12. solvent | 16. tool |



Graeme Thorne was an eight-year-old Sydney schoolboy who disappeared on his way to school one morning in July 1960, five weeks after his parents had won over \$200 000 in a lottery. His disappearance was quickly discovered when he did not meet the lady who drove him to school. The police were immediately called in. They were able to take the telephone call made by a heavily accented man demanding money in return for Graeme's safe return. Although Graeme's school case was found the day after he disappeared, his body, wrapped in a rug, was not discovered until five weeks later. Clinging to his clothing were soil, hairs and plant debris, including pollen leaves and seeds.

The rug was found to be one of 3000 manufactured between May 1955 and January 1956 by the South Australian Onkaparinga Woollen Mills. The soil on Graeme's clothes and shoes was found to be a grey clay and a pink sand, later identified by the curator of the Geological and Mining Museum to be a pink lime mortar used by builders.

The hairs on the rug were analysed by a hair expert in the forensic laboratory, and were found to be animal hair from a Pekingese or similar dog, and some bleached human hair.

The plant debris caught on the rug was identified by a botanist with the Department of Agriculture as coming from a common cypress tree (*Chamaecyparis parispasifera*) and the uncommon smooth Arizona cypress (*Cupressus glabra*). The investigating police decided they had to look for a garage at ground level, containing pink mortar, with the two types of cypress trees growing near its door.

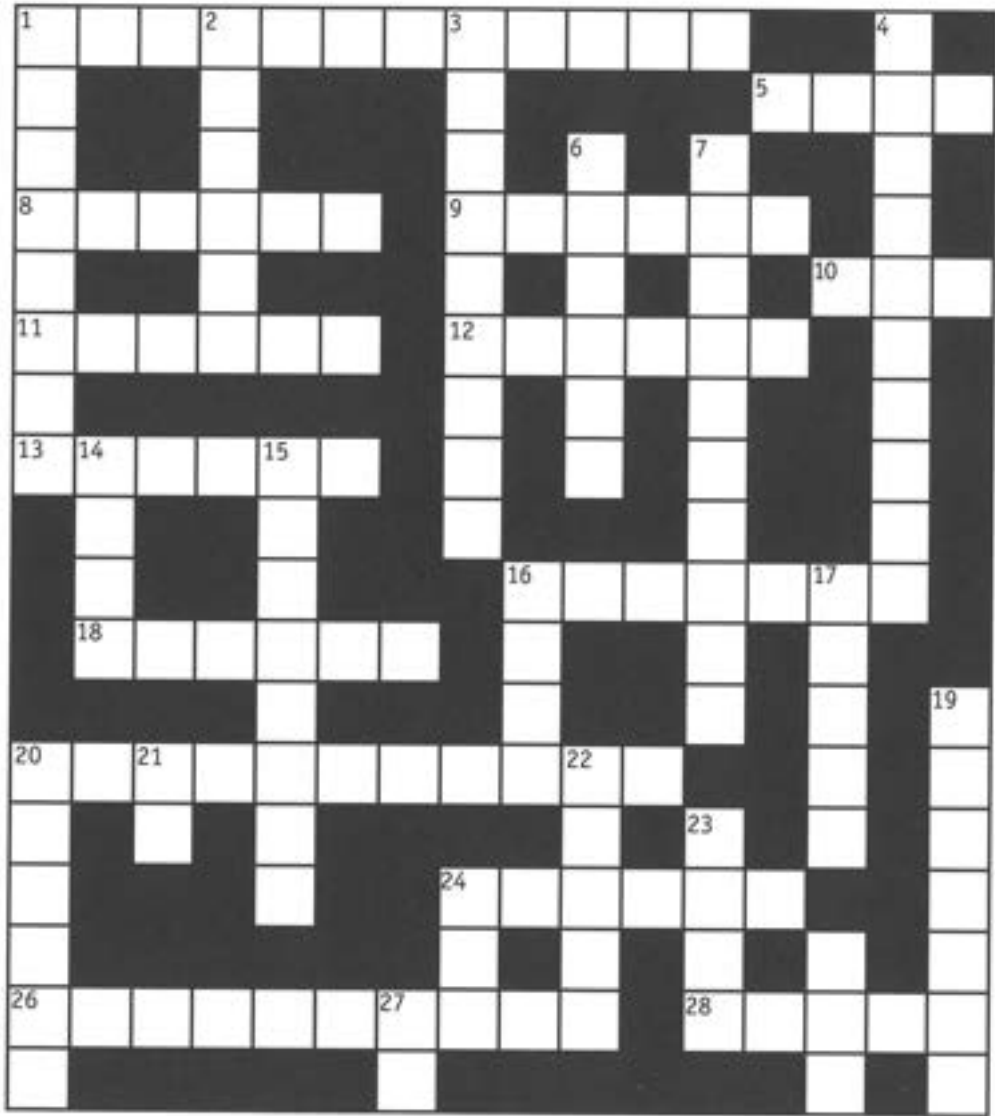
The breakthrough occurred when the police came across an amateur botanist postman who was able to show the police a place on his round where these two types of cypresses grew together. They found the two trees growing near a garage door. The garage was below the house level and all sorts of vegetation blew into it. On the floor they found some pink mortar and at the back of the garage was some grey soil, similar to that found on Graeme's clothes. There was a new owner, but the police soon found the previous occupant.

In the suspect's car police discovered similar samples of hair, dust and plant debris to that found with the boy's body. Later, string identical to that used to tie up Graeme's hands was found among the suspect's possessions. In one of the family's photograph albums was a photograph showing the family sitting on a rug identical to that found wrapped around Graeme's body. Their Pekingese dog had hair similar to that found by police, and the suspect's wife bleached her hair.

Confronted with all this evidence, the suspect confessed to the kidnapping, but not to murder. He was convicted, and died in prison of a heart attack in 1968.

? Questions

1. Make a list of all the trace contacts found with the boy's body.
2. Make a second list of all the other evidence other than trace contacts, that was also used by the investigators.
3. What evidence do you think was the most important in leading the police to the correct suspect? Explain why this was so important.
4. List three experts who analysed some of the evidence, and explain what each one tested.
5. Did all the evidence prove the suspect's guilt? Explain your answer.
6. Can you think of any evidence found in this 1960 crime that could be better analysed today? Describe how this evidence would be handled now, and in what way it would be better.
7. Even in 1960, some of the evidence could have been handled in a better way. Discuss what you think may have been done better.





Across

1. Instrument invented in the 1850s and used to produce a very clear spectrum.
5. Another name for a small flake of paint.
8. Middle layer of a hair that contains the colour.
9. Small microscopic fragments of evidence are called contact
10. Charged atom of a metal element that has its own distinct flame colour.
11. Once a paint flake or fibre is tested for flame colour, it is for further tests.
12. A forensic scientist must try to avoid these if a correct conclusion is to be reached.
13. A fibre that is produced from goat hair.
16. Outer layer of hair which consists of overlapping scales.
18. Type of microscope used for looking at paint edge-on and for sorting evidence.
20. A substance that glows under UV light is said to be
24. The colour in the visible spectrum with a wavelength longer than green.
26. The type of microscope used to compare fibres at the same time, side by side.
28. The colour in the visible spectrum that has a wavelength longer than blue light.



Down

1. The visible light spread by a prism.
2. A common natural fibre produced from a plant.
3. A type of man-made fibre.
4. An instrument used to magnify very small objects.
6. Produced when fibres are woven together.
7. The bending of light caused by light entering or leaving water or glass.
14. Liquids that are often examined under UV light in the forensic laboratory.
15. Part of electromagnetic spectrum with wavelengths longer than visible light.
16. The central part of a hair.
17. Visible has wavelengths between ultraviolet and infra-red radiation.
19. These small plant particles carry the male sex cells and survive for many years.
20. Nationality of Edmond Locard who proposed the 'Exchange Principle'.
21. Short for ultraviolet.
22. A common synthetic fibre.
23. Type of wavelength used for communication.
24. Are contact traces very important in forensic science?
25. Colour of visible light that has the longest wavelength.
27. Abbreviation for infra-red radiation.



1 Introduction to forensic science

1-1 What is forensic science?

1. Forensic science is the using of scientific techniques to solve crimes.
2. Solving crimes became a science about 100 years ago. This was influenced a lot by Locard and his work.
3. It is now more difficult for criminals because of modern techniques, and a knowledge of what evidence is useful.
4. Eyewitness reports and being caught 'red-handed'.
5. A good eyewitness is alert and observant, and notes the appearance and car registration numbers of anyone acting suspiciously.
6. This principle means that whenever there is contact between a person and another person or an object, some small particles always move from one to the other, in both directions.
7. A contact trace is the microscopic evidence that is not obvious to the untrained observer. You don't need a microscope to see a knife, but you would if there were any fingerprints on it.
8. Contact traces are important when a criminal has tried to clean up and remove all obvious evidence.

1-2 How observant are you?

1. three.
2. one, running.
3. two, one male, one female, female driving, male wearing hat.
4. late-model sedan, licence plate XYZ 123, dent and broken headlight on driver's side.
5. Collecting the mail.
6. Suburban—shops and neat houses.
7. Australia Post van.
8. Houses over the road are one-story, with TV aerials and lawns.
9. A dog running ahead of the child.
10. Boy and dog look hot and sweaty, person cycling in background, female driver looks angry/upset, male is smirking, car and postman outside fish and chip shop.
11. Description of car, reg. no., colour, damage; Direction of travel, no. of people in car and description, postman as witness and to confirm time.

1-3 Looking for differences

15 differences.

1-4 Arriving at the crime scene

1. Check for any danger to self, check any injured people, give first aid, call ambulance, do not touch anything, secure the site, call for police back-up, get names of any witnesses.

2. CSI searches the scene, detective in charge organises everything that needs to be done.
3. CSIs wear special overalls and overshoes, use duckboards, and approach the scene from the same place as the first police officer or witness did.
4. The investigators need a record of the crime scene that is as accurate as possible. Photos and maps help a court to better understand the crime.
5. A ruler in a photo helps to record the actual size of an object.
6. A pathologist is called to the scene if a body is found. Their main responsibility is to find the cause and time of death.

1-5 Removing the evidence

1. CSIs wear special overalls to prevent them from contaminating the crime scene with their own hair, etc.
2. A different CSI searches a suspect's house so there is no danger of cross-contamination of trace evidence from the crime scene to the suspect.
3. Evidence is sealed in bags so that no trace evidence is lost or contaminated in transport. It is labelled so that it cannot be misplaced.
4. Where and when (date and time) the item was found, and by whom it was collected.
5. At the forensic lab a register number is added to the label, and its movements around the lab recorded in the register so that, at any time, it is known where it is and which scientist is responsible for it.
6. If an item is contaminated, it can no longer be used as evidence in court.
7. Time and date missing from label.

1-6 The forensic science laboratory

1. (a) Dental records, age and condition of teeth to identify a body where fingerprints may not be possible, e.g. where a body is burnt beyond recognition.
(b) May give a 'profile' (characteristics) of a type of criminal who might carry out the type of crime that is being investigated.
(c) Studies insects, so may be asked to identify larvae, or the age of maggots found on a dead body to estimate the time of death.
(d) Studies body fluids, and may give an opinion on diseases, drugs or other chemicals in a victim's blood.
(e) Will inspect a dead body and do the post mortem to confirm the cause of death, etc.
2. A forensic scientist is an expert employed full-time to help solve crimes. An expert witness is anyone called in to give an expert opinion on some evidence, but may not necessarily be employed full-time in the forensic field.
3. CSIs search the crime scene for evidence, while the forensic scientist carries out tests and

- analyses the evidence scientifically.
- Order should be microscope then UV light inspection, strength and lastly melting point measurement, because the first tests should not destroy the evidence as latter tests may need to.
- Conclusion is that the two fibres are *probably* from the same garment. The rarity of the fibre would be an important factor in deciding guilt, as would any other evidence that incriminates this suspect.

1-7 Going to trial

- Jury decides the guilt in major crimes, a magistrate in minor crimes.
- The judge listens to evidence and passes sentence if the accused is found guilty.
- The defence lawyer defends the accused, trying to put forward any possibilities that support the innocence of the accused and trying to put doubt in the jury members' minds. A prosecuting lawyer must try to prove to the jury the accused's guilt beyond all reasonable doubt.
- The expert witness gives an opinion as to the probability that the accused is linked to the crime, and the evidence in which that person is an expert.
- Scientific evidence must be presented so that it is clear to all the non-scientific people in the court.
- Forensic scientists do this by using photos, videos, maps, diagrams and casts.
- To be accepted as an expert witness, a person must be recognised to have reliable and up-to-date expertise in a particular field.
- Examples of expert witnesses could include scientists at universities or other research organisations such as CSIRO or specialist doctors.
- Lawyers call in experts to demonstrate either the guilt or possible innocence of the accused, depending on which side has called in the expert.

1-9 A crime scenario: Theft at Hilltop Farm

- Tyre print in mud (plaster cast of impression), shoeprints in garden bed (plaster cast). Tool marks from screwdriver (cast made using fast-setting plasticine material). Maybe a small piece of fabric from torn jacket (bagged and later inspected under microscope, keep to compare with suspect's clothes later). Blood on broken glass (sample collected and analysed for blood group, etc. in the lab). Fingerprints on desk (dust, photograph and lift with sticky tape). Fibres from clothes left in chair he fell into (as for fabric). Paint from car on gate post (collect and analyse for car make and model), glass from broken indicator light (as for paint), oil collected from driveway (analyse to compare with oil in suspect's car).
- Pollen from wattle, glass splinters from when window was smashed, and cat hair from when he fell in the chair, could all be found caught on the suspect's clothes and link him to the location. Paint from the farm gate found on his car would also link the suspect with the crime.

- An appropriate dent on the car, a broken indicator light and the stolen medal and brooch, and the cheque book if found in the suspect's possession, may all be further evidence.
- The police would ask him where he was at the time of the theft, and to explain how all the evidence from the crime scene was on his clothes and possessions and how his car's oil, paint, glass and tyre prints were all found at the crime scene.
- The police would inspect the suspect's tools, car and clothes to try to find evidence to link the suspect with the crime.
- A general uniformed police officer would probably be called in first. They would take details of what is missing and ask the cat feeders what they found.
- If the crime was considered important, a detective would be called in to take charge of the case and a CSI would be asked to search for evidence, take photos, etc.
- The suspect may suggest that someone else had borrowed his car and committed the crime, or that he had visited the house legitimately and found that no one was home. He may say that someone gave him the brooch and medal to mind.
- The fingerprints and blood on the broken window, the glass fragments on his clothes and the tool marks on the window matching his screwdriver.

1-10 Forensic science word search



EVERY CONTACT LEAVES A TRACE.

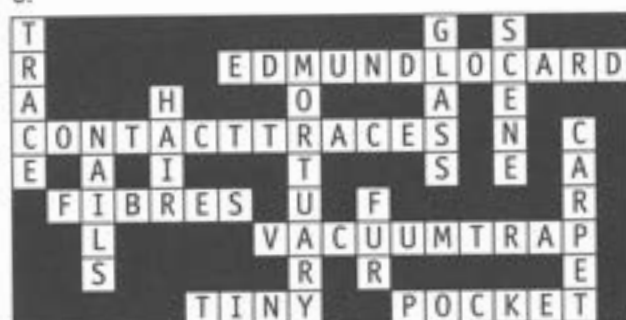
2 Contact traces

2-1 What are contact traces?

- Fingerprints** left at a crime scene can identify every individual; **hair and fibres** from the body and clothes can link the suspect to the crime; **pollens** and **glass** splinters can be removed from the crime scene on the suspect's clothes.
- Trace evidence may be caught in a **vacuum trap** if the crime scene is vacuumed; clothes and soft furnishings may have **sticky tape** pressed over them or they may be **beaten over a large funnel** to collect any particles clinging to them. A body

may have debris removed from under fingernails and clothes may be shaken above a large funnel. Blood samples will be analysed for species and blood group. In each case the evidence is bagged and labelled and sent to the laboratory for further analysis.

- The stereomicroscope is most suitable because of the large distance between the stage and the objective lens, yet it gives sufficient magnification to allow sorting of very small particles.
- The hands, feet and head of a dead body are wrapped in separate plastic bags to prevent loss of any trace evidence during transport.
- The forensic pathologist makes a thorough examination of the body in the mortuary as part of a post mortem. They look for debris under fingernails and foreign particles in air passages.



2-2 Fibres and fabrics

- Fibres are important because they can link a suspect to a crime scene. Fibres can be found in the suspect's belongings.
- Natural fibres are rougher and have a shorter staple than synthetic ones.
- Animal fibres—wool, angora, mohair, alpaca, silk, cashmere; plant fibres—cotton, linen, hemp, kapok, jute, coir, flax; synthetic—viscose, nylon, acetate, rayon, polyester, acrylic.
- Inspection under microscope, then UV light, thickness and strength measurement, then melting point. Least destructive tests are done first.
- As well as type of fibre, fabric can allow comparison of weave pattern and how it fits into a hole in a suspect's garment.

2-3 Activity: Looking at fibres

- Synthetic fibres appear smoother and more uniform.
- Colour comparison, measurement of strength, thickness and melting point.

2-4 Hair

- Hair samples may reveal a person's approximate age, race, sex and which part of the body it came from, how the hair came out, person's environment, any drugs or poisons taken or how the hair was treated (bleached, coloured, etc.).
- Scalp, face (beard, eyelashes and brows), under-arm, pubic and general body hair.
- Person's identity including their sex, race and age.
- A protein, keratin, that can only be destroyed by fire or certain chemicals.

- Inner—medulla, middle—cortex, outer—cuticle.
- Type of animal or race of person.
- The shape of the scales can indicate the type of animal from which it came.



2-5 Activity: Looking at hairs

- Hair from different parts of the body varies in thickness and in the shape of the cross-section.
- Cat, dog and rabbit hair look rougher than human hair; horse hair is thicker.
- Hairs are important Contact traces

2-6 Using light of different wavelengths

- The visible spectrum is all the light that can be seen.
- Red, Orange, Yellow, Green, Blue, Indigo, Violet.
- UV light has a shorter wavelength, IR light has a longer wavelength than visible light.
- Different objects will look different (i.e. will fluoresce differently) in UV light.
- Fluorescent is when something glows while a light is shone on it, but something that is phosphorescent will continue to glow even after the light has been removed.

2-7 Activity: Examining liquids under UV light

- Example: A case in which some oil may have been found at a crime scene and it is matched to a vehicle belonging to a suspect. Inspection and comparison of some engine oils may show up differently under UV light.

2-8 Activity: Looking at flame colours

- Colour is due to the metal part of the compound because all the compounds having the same metal have the same flame colour.
- If testing drugs or other chemicals, a forensic scientist may use flame colour or a more sophisticated instrument based on the same principle to identify the elements in an unknown substance found at a crime scene.
- A salt is a compound that is made up of a metal and a non-metal part. An ion is a charge atom, either metal or non-metal. When a salt is heated, the electrons in the metal part momentarily gain energy. However, this is quickly lost on cooling, and this energy is released in the form of coloured light. The amount of energy the electron gains is different for different metals, and this determines the unique flame colour of each metal.

- To help salts to 'stick' to the wire

2-9 Chemicals, colours and spectroscopy

- A spectroscope is a small brass tube with a prism inside it that spreads white light into a clear spectrum.
- A line spectrum of sodium has two clear yellow lines, while potassium has two bright lilac lines and two pale pink lines.
- A spectrophotometer is more accurate because it can measure the intensity of the wavelength of each sample, so it can give the amount of each element.
- An unknown substance may need to be identified in the case of suspected drugs being found.
- Advantage: it is very good at identifying elements and the amount of each in a sample.
Disadvantage: it destroys the evidence.

2-10 Glass

- Glass is made mostly from sand along with other chemicals that can give it different properties such as heat resistance, strength, brittleness or flexibility and colour. The chemicals are melted together to make the glass.
- Inspection under UV light, density and refractive index.
- Density is the amount of substance in a given volume, or how tightly particles are packed. Glass is made denser by adding lead to the sand.
- The density of two pieces of glass can be compared by floating them in a mixture of different liquids that form different layers, and have different densities. If glass floats on a layer, it is less dense than that layer; if it sinks, it is denser than that substance.
- Refractive index is a measurement of how much light is bent when passing through a transparent substance.
- The RI of two fragments of glass can be compared if they are both placed in some silicone oil which, when heated, changes its RI. If both fragments become invisible at the same time (i.e. at the same temperature), then they must have the same RI.

2-11 Activity: Glass and refractive index

- Other tests that may be carried out on the glass include inspection under UV light, comparison of density, and maybe even measurement of strength, flexibility and melting point.

2-12 Paint chips

- Inspection under microscope and solubility in different solvents.
- Conclude that the two paint flakes probably came from the same source.
- Who was driving the car at the time, any scratches or dents in appropriate places, or paint from

something found at the crime scene on the car.

- The two paint flakes may be from the same car, but from different places, which may differ in the thickness of some layers.
- Flake (c) matches flake B, so they probably came from the same car.

2-13 Activity: Looking at paint flakes

- Two identical paint flakes indicate that they probably came from the same car. This is even more likely if there are several layers on top of the original paint work. More evidence, such as dents and paint from the crime scene, would be needed to make a water-tight case.
- Dents, the suspect's whereabouts at the time of the crime and a motive would all be useful information. Evidence such as fingerprints are extremely useful.

2-14 Case study: The Graeme Thorne kidnapping

- Dog and human hair, pollen.
- The rug, plant debris (leaves and seeds), soil, string used to tie up hands, family photograph.
- The plant debris led to the kidnapper's house, and so on to him.
- Curator of Geological and Mining Museum (soil, mortar and sand), botanist with Dept. of Agriculture (plant material), forensic scientist with expertise in hairs (dog and human hair).
- The evidence did not prove the crime, but it made it extremely probable that the police had the right man.
- Nowadays fingerprints could be taken from the body and probably even from the rug.
- The police should have recorded the telephone call to match the voice demanding the ransom with a suspect.

2-15



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